Exercise 1.5

Q1:—

(a-) $\begin{bmatrix} 1 & 0 \\ -5 & 1 \end{bmatrix}$

Q1-4 (on book)

Q5:—

(a-) $E = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $A = \begin{bmatrix} -1 & -2 & 5 & -1 \\ 3 & -6 & -6 & -6 \end{bmatrix}$

$EA = \begin{bmatrix} 3 & -6 & -6 & -6 \\ -1 & -2 & 5 & -1 \end{bmatrix}$ ($R_1 \rightarrow R_2$ Interchanged)

(b-) $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix}$

$EA = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ -1 & 9 & 4 & -12 & -10 \end{bmatrix}$ $R_3 - 3R_2$

(c-) $E = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

$EA = \begin{bmatrix} -13 & 28 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$ $R_1 + 4R_3$

Question 6:—

(a-)

$E = \begin{bmatrix} -6 & 0 \\ 0 & 1 \end{bmatrix}$, $A = \begin{bmatrix} -1 & -2 & 5 & -1 \\ 3 & -6 & -6 & -6 \end{bmatrix}$

$EA = \begin{bmatrix} 6 & 12 & -30 & 6 \\ 3 & -6 & -6 & -6 \end{bmatrix}$ $\frac{-6}{6} \times R_1$

(b-)

$E = \begin{bmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $A = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ 1 & -3 & -1 & 5 & 3 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix}$

$EA = \begin{bmatrix} 2 & -1 & 0 & -4 & -4 \\ -7 & 1 & -1 & 21 & 19 \\ 2 & 0 & 1 & 3 & -1 \end{bmatrix}$ $R_2 - 4R_1$

(c-)

$$E=\begin{bmatrix}1&0&0\\0&5&0\\0&0&1\end{bmatrix}, \quad A=\begin{bmatrix}1&4\\2&5\\3&6\end{bmatrix}$$

$$EA=\begin{bmatrix}1&4\\10&25\\3&6\end{bmatrix} \quad 5\times R_2$$

Question 7:—

(a-) $EA=B$

$$A=\begin{bmatrix}3&4&1\\2&-7&-1\\8&1&5\end{bmatrix}, \quad B=\begin{bmatrix}8&1&5\\2&-7&-1\\3&4&1\end{bmatrix}$$

B was obtained for interchanging R_1, R_3 of A

So, $$E=\begin{bmatrix}0&0&1\\0&1&0\\1&0&0\end{bmatrix}$$

(b-) $EB=A$

$$B=\begin{bmatrix}8&1&5\\2&-7&-1\\3&4&1\end{bmatrix}, \quad A=\begin{bmatrix}3&4&1\\2&-7&-1\\8&1&5\end{bmatrix}$$

A is obtained from B by interchanging R_1, R_3

$$E = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

(c-) $EA = C$

$$A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 2 & -7 & 3 \end{bmatrix}$$

C is obtained for A by ~~obtaining~~ adding $-2R_1$ to R_3

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{bmatrix} \quad (R_3 - 2R_1)$$

(d-) $EC = A$

$$C = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 2 & -7 & 3 \end{bmatrix}, \quad A = \begin{bmatrix} 3 & 4 & 1 \\ 2 & -7 & -1 \\ 8 & 1 & 3 \end{bmatrix}$$

A is obtained from C by adding $2R_1$ to R_3

$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$ $(R_3 - 2R_1)$

Question : 8

(a-) $EB = D$

$$B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 8 & 1 & 5 \\ -6 & 21 & 3 \\ 3 & 4 & 1 \end{bmatrix}$$

D is obtained from B by multiplying -3 to R_2

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b-) $ED = B$

$$D = \begin{bmatrix} 8 & 1 & 5 \\ -6 & 21 & 3 \\ 3 & 4 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}$$

B is obtained from B by multiplying $-\frac{1}{3}$ to R_2

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1/3 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(c-) $EB = F$

$$B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}, \quad F = \begin{bmatrix} 8 & 1 & 5 \\ 8 & 1 & 1 \\ 3 & 4 & 1 \end{bmatrix}$$

F is obtained from B by adding $2R_1$ to R_2.

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

(d-) $EF = B$

$$F = \begin{bmatrix} 8 & 1 & 5 \\ 8 & 1 & 1 \\ 3 & 4 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 8 & 1 & 5 \\ 2 & -7 & -1 \\ 3 & 4 & 1 \end{bmatrix}$$

B is obtained from F by adding $-2R_1$ to R_2

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

Question 9:—

(a-) $A = \begin{bmatrix} 1 & 4 \\ 2 & 7 \end{bmatrix}$

$A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ By theorem 1.4.5

$= \frac{1}{(7)(1)-(4)(2)}\begin{bmatrix} 7 & -4 \\ -2 & 1 \end{bmatrix}$

$A^{-1} = \begin{bmatrix} -7 & 4 \\ 2 & -1 \end{bmatrix}$ ✓

By Inversion algorithm

$[A|I] = \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array}\right]$

$= \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 0 & -1 & -2 & 1 \end{array}\right]$ $R_2 - 2R_1$

$= \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 0 & 1 & 2 & -1 \end{array}\right]$ $-1 \times R_2$

$= \left[\begin{array}{cc|cc} 1 & 0 & -7 & 4 \\ 0 & 1 & 2 & -1 \end{array}\right]$ $R_1 - 4R_2$

$$A^{-1} = \begin{bmatrix} -7 & 4 \\ 2 & -1 \end{bmatrix}$$

(b-)

$$A = \begin{bmatrix} 2 & -4 \\ -4 & 8 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & -4 \\ -4 & 8 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 2 & -4 \\ -4 & 8 \end{vmatrix}$$

$$= 16 - 16$$

$$= 0$$

The matrix is not invertible.

Question 10:—

a) $$A = \begin{bmatrix} 1 & -5 \\ 3 & -16 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & -5 \\ 3 & -16 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & -5 \\ 3 & -16 \end{vmatrix}$$

$$= -16 + 15$$

$$= -1$$

$$A^{-1} = \frac{1}{|A|} (\text{Adj } A)$$

$$= \frac{\begin{bmatrix} -16 & 5 \\ -3 & 1 \end{bmatrix}}{-1}$$

$$= \begin{bmatrix} 16 & -5 \\ 3 & -1 \end{bmatrix}$$

By inversion algorithm:—

$$A|I = \left[\begin{array}{cc|cc} 1 & -5 & 1 & 0 \\ 3 & -16 & 0 & 1 \end{array}\right]$$

$$R_2 - 3R_1 \quad \left[\begin{array}{cc|cc} 1 & -5 & 1 & 0 \\ 0 & -1 & -3 & 1 \end{array}\right]$$

$(-1)R_2 \Rightarrow \left[\begin{array}{cc|cc} 1 & -5 & 1 & 0 \\ 0 & 1 & 3 & -1 \end{array}\right]$

$R_1 + 5R_2 \Rightarrow \left[\begin{array}{cc|cc} 1 & 0 & 16 & -5 \\ 0 & 1 & 3 & -1 \end{array}\right]$

$A^{-1} = \begin{bmatrix} 16 & -5 \\ 3 & -1 \end{bmatrix}$

b) $A = \begin{bmatrix} 6 & 4 \\ -3 & -2 \end{bmatrix}$

$A = \begin{bmatrix} 6 & 4 \\ -3 & -2 \end{bmatrix}$

$|A| = \begin{vmatrix} 6 & 4 \\ -3 & -2 \end{vmatrix}$

$= -12 + 12$

$= 0$

The matrix is not invertible.

Question 11:—

(a-) $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$

let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$

$[A^{-1}] = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array}\right]$

$R_2 - 2R_1$, $R_3 - R_1$ $= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array}\right]$

$R_3 + 2R_2 = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & +2 & +1 \end{array}\right]$

$(-1)R_3 = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$

$R_1 - 2R_2 = \left[\begin{array}{ccc|ccc} 1 & 0 & +9 & 5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$

$$\begin{matrix} R_1-9R_3 \\ R_2+3R_3 \end{matrix} \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

b) $\begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$

Let $A = \begin{bmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{bmatrix}$

$$|A| = \begin{vmatrix} -1 & 3 & -4 \\ 2 & 4 & 1 \\ -4 & 2 & -9 \end{vmatrix}$$

$$= +1(-36-2) + 3(-18+4) + 4(4+16)$$

$$= +1(-38) + 3(-14) + 4(20)$$

$$= -38-42+80$$

$$= 0$$

The matrix is not invertible.

9
-3
1

Question : 12

a) $\begin{bmatrix} 1/5 & 1/5 & -2/5 \\ 1/5 & 1/5 & 1/10 \\ 1/5 & -4/5 & 1/10 \end{bmatrix}$

$(5)R_1$, $(5)R_2$, $(5)R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 1 & 1 & 1/2 & 0 & 5 & 0 \\ 1 & -4 & 1/2 & 0 & 0 & 5 \end{array}\right]$$

$R_2 - R_1$, $R_3 - R_1$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & 0 & 5/2 & -5 & 5 & 0 \\ 0 & -5 & 5/2 & -5 & 0 & 5 \end{array}\right]$$

$R_2 \leftrightarrow R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & -5 & 5/2 & -5 & 0 & 5 \\ 0 & 0 & 5/2 & -5 & 5 & 0 \end{array}\right]$$

$\frac{-1}{5} R_2$

$$\left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & 1 & -1/2 & 1 & 0 & -1 \\ 0 & 0 & 5/2 & -5 & 5 & 0 \end{array}\right]$$

$R_1 - R_2$, $\frac{2}{5} R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -3/2 & 4 & 0 & 1 \\ 0 & 1 & -1/2 & 1 & 0 & -1 \\ 0 & 0 & 1 & -2 & 2 & 0 \end{array}\right]$$

$$R_2+\frac{3}{2}R_3,\ R_2+\frac{1}{2}R_3 \quad \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 3 & 1 \\ 0 & 1 & 0 & 0 & 1 & -1 \\ 0 & 0 & 1 & -2 & 2 & 0 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & -1 \\ -2 & 2 & 0 \end{bmatrix}$$

(b-)

$$A\,|\,I = \left[\begin{array}{ccc|ccc} 1/5 & 1/5 & -2/5 & 1 & 0 & 0 \\ 1/5 & -3/5 & -3/10 & 0 & 1 & 0 \\ 1/5 & -4/5 & 1/10 & 0 & 0 & 1 \end{array}\right]$$

$$(5)R_1,\ (5)R_2,\ (5)R_3 \quad \left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 2 & -3 & -3/2 & 0 & 5 & 0 \\ 1 & -4 & 1/2 & 0 & 0 & 5 \end{array}\right]$$

$$R_2-2R_3,\ R_3-R_1 \quad \left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & -5 & 5/2 & -10 & -5 & 0 \\ 0 & -5 & 5/2 & -5 & 0 & 5 \end{array}\right]$$

$$R_3-R_2 \quad \left[\begin{array}{ccc|ccc} 1 & 1 & -2 & 5 & 0 & 0 \\ 0 & -5 & 5/2 & -10 & 5 & 0 \\ 0 & 0 & 0 & 5 & -5 & 5 \end{array}\right]$$

Since a

side so

Qu

$A|I = \left[\begin{array}{c} 1 \\ 0 \\ 1 \end{array}\right.$

$R_3-R_1 \left[\begin{array}{c} 1 \\ 0 \\ 0 \end{array}\right.$

$R_3-R_2 \left[\begin{array}{c} 1 \\ 0 \\ 0 \end{array}\right.$

$-\frac{1}{2}R_3 \left($

R_1-R_3

R_2-R_3

Since a row of zero is obtained on left side so A is not invertible.

Question 13 :—

$$A = A|I = \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$R_3 - R_1 \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & -1 & -1 & 0 & 1 \end{array}\right]$$

$$R_3 - R_2 \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & -2 & -1 & -1 & 1 \end{array}\right]$$

$$-\frac{1}{2} R_3 \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1/2 & 1/2 & 1/2 \end{array}\right]$$

$$\begin{matrix} R_1 - R_3 \\ R_2 - R_3 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & -1/2 & 1/2 \\ 0 & 1 & 0 & -1/2 & 1/2 & 1/2 \\ 0 & 0 & 1 & 1/2 & 1/2 & -1/2 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ -1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 \end{bmatrix}$$

Question 14:—

$$A|I = \left[\begin{array}{ccc|ccc} \sqrt{2} & 3\sqrt{2} & 0 & 1 & 0 & 0 \\ -4\sqrt{2} & \sqrt{2} & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\left(\frac{1}{\sqrt{2}}\right)R_1, \left(\frac{1}{\sqrt{2}}\right)R_2 \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1/\sqrt{2} & 0 & 0 \\ -4 & 1 & 0 & 0 & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$R_2 + 4R_1 \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1/\sqrt{2} & 0 & 0 \\ 0 & 13 & 0 & 4/\sqrt{2} & 1/\sqrt{2} & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$\frac{1}{13}R_2 \left[\begin{array}{ccc|ccc} 1 & 3 & 0 & 1/\sqrt{2} & 0 & 0 \\ 0 & 1 & 0 & 2\sqrt{2}/13 & 1/13\sqrt{2} & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$R_1 - 3R_2 \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 5\sqrt{2}/26 & -3\sqrt{2}/26 & 0 \\ 0 & 1 & 0 & 2\sqrt{2}/13 & 1/13\sqrt{2} & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

$$= \begin{bmatrix} \sqrt{2}/26 & -3\sqrt{2}/26 & 0 \\ 2\sqrt{2}/13 & \frac{1}{13\sqrt{2}} & \frac{\sqrt{2}}{26} \\ 0 & 0 & 1 \end{bmatrix}$$

Question 15:—

$$A=[A|I] = \left[\begin{array}{ccc|ccc} 2 & 6 & 6 & 1 & 0 & 0 \\ 2 & 7 & 6 & 0 & 1 & 0 \\ 2 & 7 & 7 & 0 & 0 & 1 \end{array}\right]$$

$$\left(\frac{1}{2}R_1, R_2, R_3\right) \left[\begin{array}{ccc|ccc} 1 & 3 & 3 & 1/2 & 0 & 0 \\ 1 & 7/2 & 3 & 0 & 1/2 & 0 \\ 1 & 7/2 & 7/2 & 0 & 0 & 1/2 \end{array}\right]$$

$$\begin{matrix} R_2-R_1 \\ R_3-R_1 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 3 & 3 & 1/2 & 0 & 0 \\ 0 & 1/2 & 0 & -1/2 & 1/2 & 0 \\ 0 & 1/2 & 1/2 & -1/2 & 0 & 1/2 \end{array}\right]$$

$$\begin{matrix} 2R_2 \\ 2R_3 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 3 & 3 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 1 & 1 & -1 & 0 & 1 \end{array}\right]$$

$$\begin{matrix} R_1-3R_2 \\ R_3-R_1 \end{matrix} \left[\begin{array}{ccc|ccc} 1 & 0 & 3 & 7/2 & -3 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array}\right]$$

$R_1 - 3R_3$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 7/2 & 0 & -3 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{array}\right]$$

$$= \begin{bmatrix} 7/2 & 0 & -3 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

Question 16:—

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 1 & 3 & 5 & 0 \\ 1 & 3 & 5 & 7 \end{bmatrix}$$

let $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 1 & 3 & 5 & 0 \\ 1 & 3 & 5 & 7 \end{bmatrix}$

Expand by C_4

$$(7)^{****} \times \begin{vmatrix} 1 & 3 & 0 \\ 1 & 3 & 5 \\ 1 & 3 & 5 \end{vmatrix}$$

two identical rows or two identical column, their

$|A| = 0$. The matrix is not invertible.

Question 17:-

$$\begin{bmatrix} 2 & -4 & 0 & 0 \\ 1 & 2 & 12 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & -1 & -4 & -5 \end{bmatrix}$$

let $A = \begin{bmatrix} 2 & -4 & 0 & 0 \\ 1 & 2 & 12 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & -1 & -4 & -5 \end{bmatrix}$

Expand by C_4

$$= (+5)^{4+4} \begin{vmatrix} 2 & -4 & 0 \\ 1 & 2 & 12 \\ 0 & 0 & 2 \end{vmatrix}$$

Expand by R_3

$$= (+5)(-2)^{3+3}(4+4)$$

$$= -80$$

$$\left[\begin{array}{cccc|cccc} 2 & -4 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & -4 & -5 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_1 \leftrightarrow R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 2 & -4 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & -4 & -5 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_2 - 2R_1$

$$\left[\begin{array}{cccc|cccc} 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 0 & -8 & -24 & 0 & 1 & -2 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & -4 & -5 & 0 & 0 & 0 & 1 \end{array}\right]$$

$\left(\frac{-1}{8}\right)R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & -1/8 & 1/4 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & -4 & -5 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_4 + R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & -1/8 & 1/4 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & -5 & -1/8 & 1/4 & 0 & 1 \end{array}\right]$$

$\left(\frac{1}{2}\right)R_3$

$R_4 + R_3$

$\left(\frac{1}{-5}\right)R_4$

$R_3 - 2R$

R_2
R

$$\left(\tfrac{1}{2}\right)R_3 \begin{bmatrix} 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & -1/8 & 1/4 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & -1 & -5 & -1/8 & 1/4 & 0 & 1 \end{bmatrix}$$

$$R_4 + R_3 \begin{bmatrix} 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & -1/8 & 1/4 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & -5 & -1/8 & 1/4 & 1/2 & 1 \end{bmatrix}$$

$$\left(\tfrac{1}{-5}\right)R_4 \begin{bmatrix} 1 & 2 & 12 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 3 & 0 & -1/8 & 1/4 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1 & 1/40 & -1/20 & -1/10 & -1/5 \end{bmatrix}$$

$$R_1 - 2R_2 \begin{bmatrix} 1 & 0 & 6 & 0 & 1/4 & 1/2 & 0 & 0 \\ 0 & 1 & 3 & 0 & -1/8 & 1/4 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1 & 1/40 & -1/20 & -1/10 & -1/5 \end{bmatrix}$$

$$\begin{matrix} R_1 - 6R_3 \\ R_2 - 3R_3 \end{matrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 1/4 & 1/2 & -3 & 0 \\ 0 & 1 & 0 & 0 & -1/8 & 1/4 & -3/2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 1 & 1/40 & -1/20 & -1/10 & -1/5 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 1/4 & 1/2 & -3 & 0 \\ -1/8 & 1/4 & -3/2 & 0 \\ 0 & 0 & 1/2 & 0 \\ 1/40 & -1/20 & -1/10 & -1/5 \end{bmatrix}$$

Question 18:—

$$A|I = \left[\begin{array}{cccc|cccc} 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 3 & 0 & 0 & 0 & 1 & 0 \\ 2 & 1 & 5 & -3 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_1 \leftrightarrow R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & -1 & 3 & 0 & 0 & 0 & 1 & 0 \\ 2 & 1 & 5 & -3 & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_4 - 2R_2$, $(-1)R_3$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & -3 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 5 & -5 & 0 & -2 & 0 & 1 \end{array}\right]$$

$R_2 \leftrightarrow R_3$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -3 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 5 & -5 & 0 & -2 & 0 & 1 \end{array}\right]$$

$R_4 - R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -3 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 8 & -5 & 0 & -2 & 1 & 1 \end{array}\right]$$

$\frac{1}{2} R_3$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -3 & 0 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 8 & -5 & 0 & -2 & 1 & 1 \end{array}\right]$$

$R_2 + 3R_3$
$R_4 - 8R_3$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 3/2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 0 & -5 & -4 & -2 & 1 & 1 \end{array}\right]$$

$-\frac{1}{5} R_4$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 3/2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 4/5 & 2/5 & -1/5 & -1/5 \end{array}\right]$$

$R_1 - R_4$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & -4/5 & 3/5 & 1/5 & 1/5 \\ 0 & 1 & 0 & 0 & 3/2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 1/2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 4/5 & 2/5 & -1/5 & -1/5 \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} -4/5 & 3/5 & 1/5 & 1/5 \\ 3/2 & 0 & -1 & 0 \\ 1/2 & 0 & 0 & 0 \\ 4/5 & 2/5 & -1/5 & -1/5 \end{bmatrix}$$

Question 19:—

(a-) $\begin{bmatrix} k_1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 \\ 0 & 0 & k_3 & 0 \\ 0 & 0 & 0 & k_4 \end{bmatrix}$

let $A = \begin{bmatrix} k_1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 \\ 0 & 0 & k_3 & 0 \\ 0 & 0 & 0 & k_4 \end{bmatrix}$

$$A^{-1} = \left[\begin{array}{cccc|cccc} k_1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & k_3 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & k_4 & 0 & 0 & 0 & 1 \end{array}\right]$$

$$\left(\frac{1}{k_1}\right)R_1 \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k_1 & 0 & 0 & 0 \\ 0 & k_2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & k_3 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & k_4 & 0 & 0 & 0 & 1 \end{array}\right]$$

$\left(\frac{1}{k_2}\right)R_2$ $\begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$\left(\frac{1}{k_3}\right)R_3$ $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$\left(\frac{1}{k_4}\right)R_4$

$A^{-1} =$

(b-)

$A^{-1} =$

$$\left(\tfrac{1}{k}\right)R_1 \quad \left(\begin{array}{cccc|cccc} 1 & 1/k & 0 & 0 & 1/k & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1/k & 0 & 0 & 1/k & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right)$$

$(1/k)R_3$

$$R_1 - \tfrac{1}{k}R_2 = \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & -1/k & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1/k & 0 & 0 & 1/k & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right)$$

$\frac{1}{k} - \frac{1}{k}(*)$

$\frac{1}{k} - \frac{1}{k}$

$\frac{1}{k} - \frac{1}{k} = 0$

$$R_3 - \tfrac{1}{k}R_4 \quad \left(\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & -1/k & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1/k & -1/k \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$= \left(\begin{array}{cccc} 1/k & -1/k & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1/k & -1/k \\ 0 & 0 & 0 & 1 \end{array}\right)$$

Question 20

a)

$$\left[\begin{array}{cccc|cccc} 0 & 0 & 0 & k_1 & 1 & 0 & 0 & 0 \\ 0 & 0 & k_2 & 0 & 0 & 1 & 0 & 0 \\ 0 & k_3 & 0 & 0 & 0 & 0 & 1 & 0 \\ k_4 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right]$$

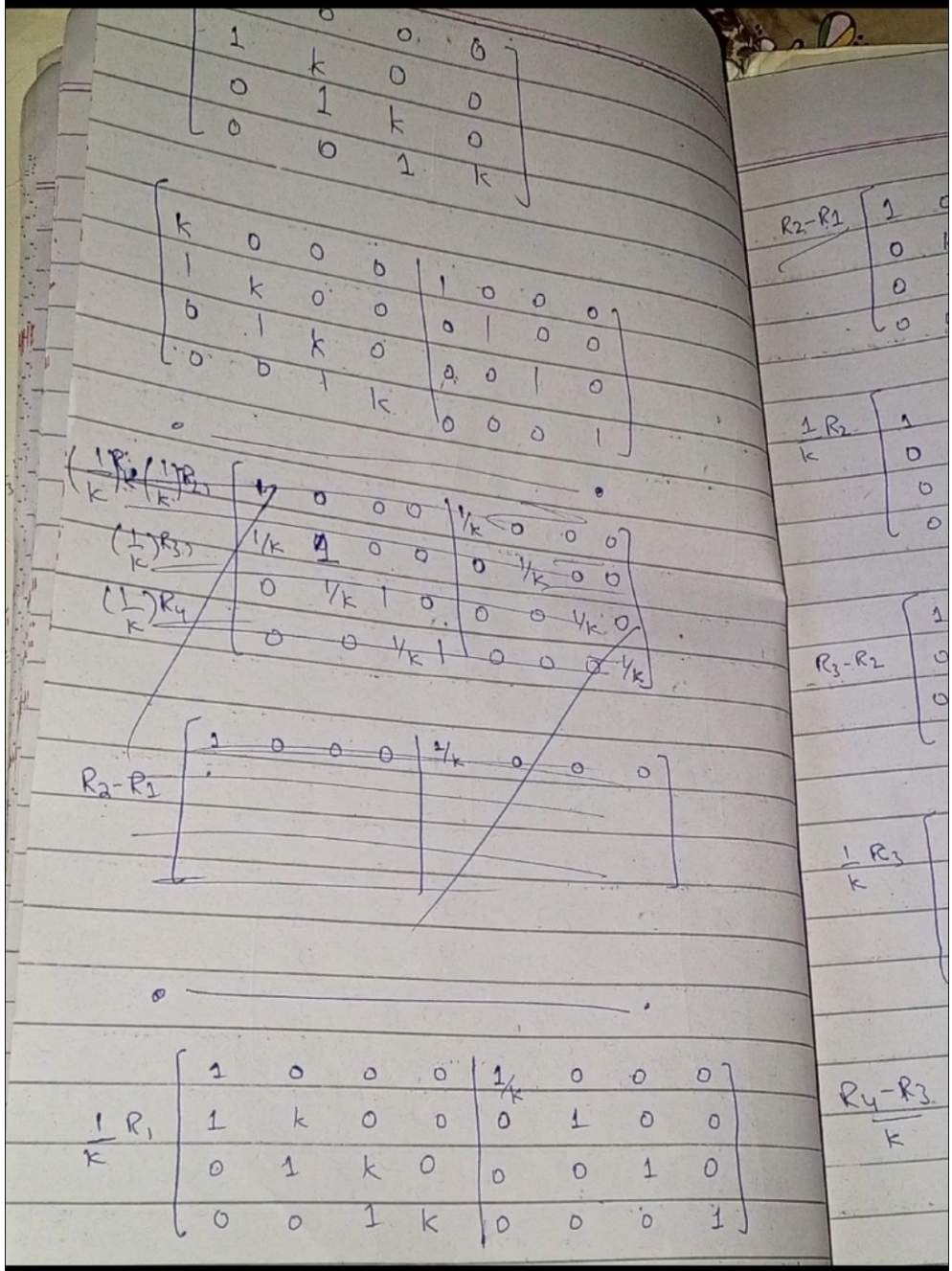
Interchange by c_1 & c_4 & c_2 & c_3

$$\left[\begin{array}{cccc|cccc} k_1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & k_2 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & k_3 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & k_4 & 1 & 0 & 0 & 0 \end{array}\right]$$

$$\begin{matrix} \left(\frac{1}{k_1}\right)R_1 \\ \left(\frac{1}{k_2}\right)R_2 \\ \left(\frac{1}{k_3}\right)R_3 \\ \left(\frac{1}{k_4}\right)R_4 \end{matrix} \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1/k_1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1/k_2 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1/k_3 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1/k_4 & 0 & 0 & 0 \end{array}\right]$$

$$A^{-1} = \left[\begin{array}{cccc} 0 & 0 & 0 & 1/k_1 \\ 0 & 0 & 1/k_2 & 0 \\ 0 & 1/k_3 & 0 & 0 \\ 1/k_4 & 0 & 0 & 0 \end{array}\right]$$

(b-)



$R_2 - R_1$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & 0 & 0 & 0 \\ 0 & k & 0 & 0 & -1/k & 1 & 0 & 0 \\ 0 & 1 & k & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & k & 0 & 0 & 0 & 1 \end{array}\right]$$

$\frac{1}{k} R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & 0 & 0 & 0 \\ 0 & 1/k & 0 & 0 & -1/k^2 & 1/k & 0 & 0 \\ 0 & 1 & k & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & k & 0 & 0 & 0 & 1 \end{array}\right]$$

$R_3 - R_2$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1/k^2 & 1/k & 0 & 0 \\ 0 & 0 & k & 0 & 0+1/k^2 & 0-1/k & 1 & 0 \\ 0 & 0 & 1 & k & 0 & 0 & 0 & 1 \end{array}\right]$$

$\frac{1}{k} R_3$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1/k & 1/k & 0 & 0 \\ 0 & 0 & 1 & 0 & \frac{1/k^2}{k} & \frac{-1/k}{k} & 1/k & 0 \\ 0 & 0 & 1 & k & 0 & 0 & 0 & 1 \end{array}\right]$$

$\frac{R_4 - R_3}{k}$

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1/k & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1/k^2 & 1/k & 0 & 0 \\ 0 & 0 & 1 & 0 & 1/k^3 & -1/k^2 & 1/k & 0 \\ 0 & 0 & 0 & 1 & -1/k^4 & 1/k^3 & -1/k^2 & 1/k \end{array}\right]$$

$$A^{-1} = \begin{bmatrix} 1/k & 0 & 0 & 0 \\ -1/k^2 & 1/k & 0 & 0 \\ -1/k^3 & 1/k^2 & 1/k & 0 \\ 1/k^4 & +1/k^3 & -1/k^2 & 1/k^1 \end{bmatrix}$$

Q 21:—

$$\begin{bmatrix} c & c & c \\ 1 & c & c \\ 1 & 1 & c \end{bmatrix}$$

$$c(c^2-c) - c(c-c) + c(1-c)$$

$$c(c^2-c) - 0 + c(1-c)$$

$$c^3 - c^2 + c - c^2$$

$$c^3 - 2c^2 + c$$

$$c(c^2 - 2c + 1)$$

$$c(c-1)^2$$

$|A| = 0$ at $c = 0$ & $c = 1$

A is not invertible at $c = 0$ as $c = 1$ & A is invertible for all values of c except 0 & 1

Q22:

$$\begin{bmatrix} c & 1 & 0 \\ 1 & c & 1 \\ 0 & 1 & c \end{bmatrix}$$

$c(c^2-1) - 1(c-0) + 0(1-0)$

$c(c^2-1) - 1(c) + 0$

$c^3 - c - c + 0$

$c^3 - 2c$

$c(c^2-2)$

$2(4-2)$
$2(2)$
4

$c = \pm\sqrt{2}$, $c = 0$

~~$|A| = 0$ at $c=0$ & $c=2$. $\neq 0$, A~~

$|A| = 0$ is not invertible at $c = 0$ as $c = 2$ & A is invertible for all values of c except 0 & 2.